Lightning node: start automatically on the Mac

Altradits LND node and ngrok tunnel as macOS launch agents. Set up 8 November 2026.

What this does

Two launch agents start your Lightning node (LND) and the ngrok tunnel when you log in to the Mac. If either one crashes, macOS starts it again. You no longer start them by hand after a reboot, and they keep running when you close Terminal.

Cost: nothing. Launch agents are built into macOS. No account, no payment and no internet are needed to set them up. ngrok stays on its free plan and free domain.

Internet: the node and the tunnel still need internet to sync and to take payments. If the Mac is offline they wait, and they reconnect by themselves when the internet returns.

Tor already starts by itself (Homebrew's own launch agent, sh.brew.tor), so it needs nothing here.

The two files

File	Starts	Restarts it when
`~/Library/LaunchAgents/com.altradits.lnd.plist`	LND node	it crashes
`~/Library/LaunchAgents/com.altradits.ngrok.plist`	ngrok tunnel	it stops for any reason

Logs go to `~/.lnd-altradits/logs/launchd-lnd.log` and `launchd-ngrok.log`.

Setup, step by step

Run each block in Terminal. Steps 1 to 6 are already done on this Mac. Use them to set it up again on a new Mac, or after undoing it.

Step 1. Create the node's launch agent

Save this as `~/Library/LaunchAgents/com.altradits.lnd.plist`. *KeepAlive* / *SuccessfulExit false* means macOS restarts LND after a crash, but not after you stop it on purpose.

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>com.altradits.lnd</string>
  <key>ProgramArguments</key>
  <array>
    <string>/Users/you/bin/lnd</string>
    <string>--lnddir=/Users/you/.lnd-altradits</string>
  </array>
  <key>RunAtLoad</key>
  <true/>
  <key>KeepAlive</key>
  <dict>
    <key>SuccessfulExit</key>
    <false/>
  </dict>
  <key>ThrottleInterval</key>
  <integer>30</integer>
  <key>StandardOutPath</key>
  <string>/Users/you/.lnd-altradits/logs/launchd-lnd.log</string>
  <key>StandardErrorPath</key>
  <string>/Users/you/.lnd-altradits/logs/launchd-lnd.log</string>
</dict>
</plist>
```

Step 2. Create the tunnel's launch agent

Save this as `~/Library/LaunchAgents/com.altradits.ngrok.plist`. It uses your existing ngrok-policy.yml, so only the website server's IP can reach the node.

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>com.altradits.ngrok</string>
  <key>ProgramArguments</key>
  <array>
    <string>/opt/homebrew/bin/ngrok</string>
    <string>http</string>
    <string>https://localhost:8080</string>
    <string>--url=https://[your-domain].ngrok-free.dev</string>
    <string>--traffic-policy-file</string>
    <string>/Users/you/.lnd-altradits/ngrok-policy.yml</string>
    <string>--log=stdout</string>
  </array>
  <key>RunAtLoad</key>
  <true/>
  <key>KeepAlive</key>
  <true/>
  <key>ThrottleInterval</key>
  <integer>30</integer>
  <key>StandardOutPath</key>
  <string>/Users/you/.lnd-altradits/logs/launchd-ngrok.log</string>
  <key>StandardErrorPath</key>
  <string>/Users/you/.lnd-altradits/logs/launchd-ngrok.log</string>
</dict>
</plist>
```

Step 3. Check both files are valid

```
plutil -lint ~/Library/LaunchAgents/com.altradits.*.plist
```

Both lines must end in **OK**.

Step 4. Stop any copies you started by hand

Only one LND can run at a time. A second copy fails with *bind: address already in use* and *exit 1*.

```
pkill -TERM -x lnd      # LND shuts down cleanly; wait about 10 seconds
pkill -x ngrok
```

Step 5. Load the launch agents

```
launchctl bootstrap gui/$(id -u) ~/Library/LaunchAgents/com.altradits.lnd.plist
launchctl bootstrap gui/$(id -u) ~/Library/LaunchAgents/com.altradits.ngrok.plist
launchctl list | grep altradits
```

Each line shows a number (the process ID), then 0, then the name. From now on they also load by themselves every time you log in.

Step 6. Check the node and the tunnel

Wait about a minute, then:

```
~/bin/lncli --lnddir=$HOME/.lnd-altradits --network=mainnet \
  --macaroonpath=$HOME/.lnd-altradits/data/chain/bitcoin/mainnet/readonly.macaroon getinfo \
  | grep -E '"synced_to_chain"|"num_active_channels"'

curl -s http://127.0.0.1:4040/api/tunnels | grep -o '"public_url":"[^"]*"'
```

Ready when **synced_to_chain** is **true**, **num_active_channels** is **1**, and the second command prints the ngrok-free.dev address.

Step 7. Delete the admin key after every start

Each time LND starts it creates admin.macaroon again, the key with full control of the node. The website only uses the receive-only invoice key, so delete the admin key:

```
rm ~/.lnd-altradits/data/chain/bitcoin/mainnet/admin.macaroon
```

Do this after each reboot or restart. For day-to-day checks use readonly.macaroon (your lncli-ro alias).

Everyday commands

Restart the node	`launchctl kickstart -k gui/$(id -u)/com.altradits.lnd`
Restart the tunnel	`launchctl kickstart -k gui/$(id -u)/com.altradits.ngrok`
Is it running?	`launchctl list | grep altradits`
Read the logs	`tail -50 ~/.lnd-altradits/logs/launchd-lnd.log` `tail -50 ~/.lnd-altradits/logs/launchd-ngrok.log`

After a node restart, do step 7 again.

If something goes wrong

bind: address already in use / exit 1: another LND is already running. Nothing is broken. Check with `pgrep -fl lnd`.

Website shows ERR_NGROK_3200: the tunnel is offline. Check the Mac is online, then restart the tunnel.

Opening the ngrok address from your own browser is blocked: that is expected. The policy only lets the website server ([website server IP]) in.

Mac asleep: the node can't take payments while the Mac sleeps. Keep it plugged in and set System Settings > Energy to stop it sleeping when the display is off.

To undo

```
launchctl bootout gui/$(id -u)/com.altradits.lnd
launchctl bootout gui/$(id -u)/com.altradits.ngrok
rm ~/Library/LaunchAgents/com.altradits.lnd.plist
rm ~/Library/LaunchAgents/com.altradits.ngrok.plist
```

This stops both and removes auto-start. Your wallet, channels and funds are not touched.